

SKILL MATCHER

This project addresses inefficiencies in hiring by leveraging AI to match skills between resume and job descriptions.

Outcome of the project: Research paper

PCSE25-57

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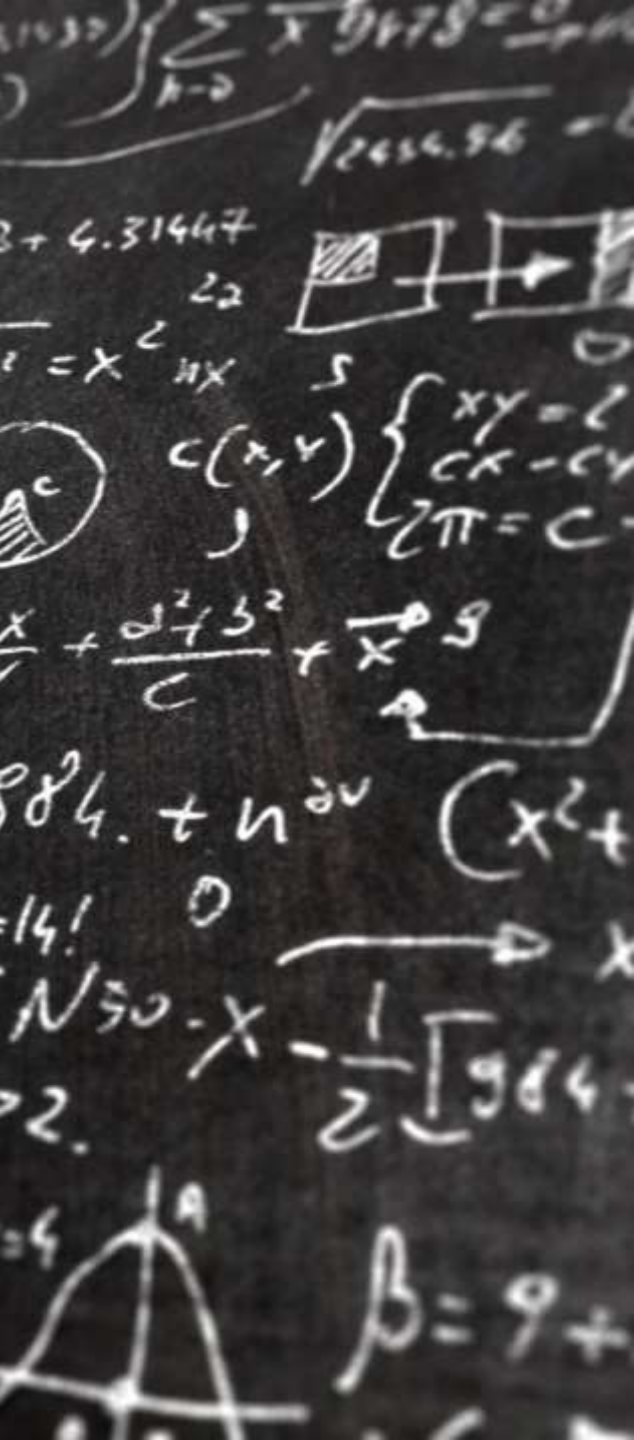
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Course Outcome	Sustainable Development Goals
CO1: To analyse and describe the problem domain	SDG 8: Decent Work and Economic Growth
CO2: To formulate clear work plan and procedure	SDG 9: Industry, Innovation & Infrastructure
CO3: To describe and evaluate both generic and specific skills	SDG 4: Quality Education
CO4: To design and apply modern tools	SDG 10: Reduced Inequality
CO5: To design report and presentation	SDG 17: Partnerships for the Goals

A vertical strip on the left side of the slide shows a chalkboard with various handwritten mathematical notes. These include a summation formula $\sum_{h=0}^{\infty} x^h$, a square root $\sqrt{2436.96}$, a fraction $\frac{3+4.31447}{22}$, a function $c(x,y)$ with a system of equations $\begin{cases} xy=2 \\ cx-cy \end{cases}$ and $2\pi=c$, a complex fraction $\frac{x}{c} + \frac{d^2 + 3^2}{c} + \frac{1}{x^2}$, a term $884. + n^{20}$, a factorial $14!$, a term $N^{30} \cdot x$, a fraction $\frac{1}{2} \sqrt{984}$, and a graph of a parabola with points A and B. The text $\beta = 9 +$ is also visible.

COURSE OUTCOME – SUSTAINABLE DEVELOPMENT GOALS

- CO1: TO ANALYZE AND DESCRIBE THE PROBLEM DOMAIN –
SDG 8: DECENT WORK AND ECONOMIC GROWTH
- CO2: TO FORMULATE CLEAR WORK PLAN AND PROCEDURE –
SDG 9: INDUSTRY, INNOVATION & INFRASTRUCTURE
- CO3: TO DESCRIBE AND EVALUATE BOTH GENERIC AND
SPECIFIC SKILLS – SDG 4: QUALITY EDUCATION
- CO4: TO DESIGN AND APPLY MODERN TOOLS – SDG 10:
REDUCED INEQUALITY
- CO5: TO DESIGN REPORT AND PRESENTATION – SDG 17:
PARTNERSHIPS FOR THE GOALS

PROJECT ABSTRACT

Overview

THIS PROJECT AUTOMATES SKILL EXTRACTION AND JOB-RESUME MATCHING USING A FINE-TUNED BERT-BASED NLP MODEL. IT PROVIDES EMPLOYERS AND CANDIDATES WITH ANALYTICS AND REAL-TIME MARKET TRENDS TO IMPROVE HIRING EFFICIENCY AND DECISION-MAKING.

Key Objectives

To **automate and optimize the recruitment process** by accurately extracting and matching skills from resumes and job descriptions using advanced AI techniques, primarily a **BERT-based NLP model**.

Approach

The **approach** of the **Skill Matcher** project combines advanced Natural Language Processing (NLP), Machine Learning (ML), and data engineering to develop an automated and intelligent job-resume matching system.

PROJECT GOALS AND OBJECTIVES



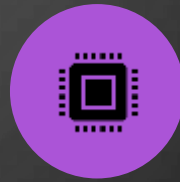
1. AUTOMATED
RESUME SCREENING
USING BERT-BASED
NER.



2. ACCURATE SKILL
IDENTIFICATION WITH
SEMANTIC SIMILARITY
METRICS.



3. MARKET TREND
ANALYTICS BASED ON
JOB LOCATION,
DOMAIN, SALARY.



4. USER-FRIENDLY
INTERFACE WITH REAL-
TIME JOB
SUGGESTIONS.



5. SCALABLE
ARCHITECTURE USING
MODULAR PIPELINES
AND DATABASES.

METHODOLOGY AND APPROACH



System Architecture Design:
Transformer-based model + Graph
DB



Data Sources: LinkedIn Scraper +
Kaggle Datasets



NER using BERT: Custom BIO tagging
schema



Visualization Tools: Neo4j +
Dashboard for live insights

EXPECTED OUTCOMES AND IMPACT

- - ENHANCED PRECISION IN SKILL DETECTION
- - LIVE JOB-MARKET ANALYTICS
- - PERSONALIZED CAREER MATCHING
- - STRATEGIC WORKFORCE PLANNING TOOLS
- - 81% RESUME-JOB MATCH ACCURACY

KEY DELIVERABLES

- - SKILL EXTRACTION ENGINE: NER USING BERT
- - ANALYTICS DASHBOARD: LIVE TREND GRAPHS, TOP SKILLS, AND HIRING DEMAND
- - RESUME MATCHING PLATFORM: SEMANTIC COMPARISON ENGINE USING EMBEDDINGS
- - SCALABLE DATA PIPELINE: MODULAR INGESTION, PROCESSING, AND VISUALIZATION

NEXT STEPS AND CONCLUSION



1. Multilingual Support for non-English resumes/job posts



2. Real-time Integration with external APIs



3. Behavioral Assessments and explainable AI models



4. Government and Industry Collaboration for broader deployment